Early Disease Detection from Lifestyle Data

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Dataset: `Wellbeing_and_lifestyle_data_Kaggle.csv` (path: `/content/sample_data/Wellbeing_and_lifestyle_data_Kaggle.csv`)

Abstract

This report documents an analysis and modeling pipeline built to detect early signs of disease using lifestyle data from the provided Kaggle dataset. The work covers data understanding, cleaning, exploratory data analysis (EDA), feature engineering, model selection and training, evaluation with appropriate metrics, and discussion of results. The goal is to build and evaluate models that can predict disease risk from lifestyle indicators and to provide recommendations for future improvements and deployment.

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1. Introduction

Lifestyle choices play a major role in the onset and progression of many chronic diseases. Using behavioral and demographic data, machine learning models can identify individuals at elevated risk and enable early interventions. This project analyzes a wellbeing and lifestyle dataset to detect early disease indicators and evaluates several predictive models.

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2. Objectives

- Understand the structure and quality of the dataset.
- Perform EDA to highlight important features and patterns.
- Engineer features suitable for predictive modeling.
- Train and compare multiple classification models for disease detection.
- Evaluate models using metrics relevant to imbalanced classification (precision, recall, F1-score, ROC-AUC).
- Provide recommendations and next steps for model improvement and deployment.

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3. Dataset Description

Source: Kaggle — `Wellbeing_and_lifestyle_data_Kaggle.csv`

Path used in project:
`/content/sample_data/Wellbeing_and_lifestyle_data_Kaggle.csv`

Typical columns (examples — update to exact names from the notebook):

- `age` — participant age
- `gender` — gender
- `physical_activity` — activity level / minutes per week
- `sleep_hours` — average sleep hours per night
- `diet_score` — measured diet quality
- `alcohol_consumption` — frequency or quantity
- `smoking_status` — smoker / non-smoker
- `stress_level` — self-reported stress
- `chronic_condition` / `disease_label` — target variable indicating disease presence or high risk

4. Methodology

4.1 Data cleaning & preprocessing

```
Missing Values:
FRUITS_VEGGIES            0
DAILY_STRESS              0
PLACES_VISITED            0
CORE_CIRCLE               0
SUPPORTING_OTHERS         0
SOCIAL_NETWORK            0
ACHIEVEMENT               0
DONATION                  0
BMI_RANGE                 0
TODO_COMPLETED            0
FLOW                      0
DAILY_STEPS               0
LIVE_VISION               0
SLEEP_HOURS               0
LOST_VACATION             0
DAILY_SHOUTING            0
SUFFICIENT_INCOME         0
PERSONAL_AWARDS           0
TIME_FOR_PASSION          0
WEEKLY_MEDITATION         0
AGE                       0
GENDER                    0
WORK_LIFE_BALANCE_SCORE   0
disease_risk              0
dtype: int64

Class Distribution:
disease_risk
0    7989
1    7983
Name: count, dtype: int64
```

Key steps performed (adjust to what your notebook actually does):

- Remove duplicate rows: `df.drop_duplicates()`
- Handle missing values: strategies used — drop, impute (median for numeric, mode for categorical), or flag missingness.
- Convert datatypes: ensure numeric columns are numeric, categorical columns as `category`.
- Encode categorical variables: one-hot encoding or ordinal encoding where appropriate.
- Scale numeric features: standard scaler or min-max scaler for algorithms sensitive to scale.

4.2 Exploratory Data Analysis (EDA)

EDA steps to include in the report (summarize the notebook outputs here):

- Distribution plots of key features (histograms, boxplots)
- Correlation matrix and heatmap to find relationships between predictors and target
- Class balance check for the target variable
- Grouped statistics (e.g., mean diet_score by disease status)

4.3 Feature Engineering

Typical operations:

- Derive BMI from weight and height if available
- Aggregate physical activity features into a single score
- Create binary flags (e.g., `is_smoker`, `high_alcohol`) from frequency columns
- Interaction terms (e.g., `age * physical_activity`)

Feature selection:

- Use correlation thresholding, univariate selection (Chi-squared), or model-based importance (Random Forest) to reduce dimensionality if needed.

4.4 Modeling approach

Algorithms trained (typical choices; update to actual models used in notebook):

- Logistic Regression (baseline)
- Random Forest Classifier
- XGBoost / Gradient Boosting Classifier
- Support Vector Machine (optional)

Train/test split & validation:

- Hold-out split e.g., `train_test_split(df, test_size=0.2, stratify=y)`
- Cross-validation (k-fold, stratified) for robust performance estimates
- Use grid search or randomized search for hyperparameter tuning

4.6 Evaluation strategy

Evaluation metrics used (especially for imbalanced classes):

- Accuracy (informative if classes balanced)
- Precision
- Recall (sensitivity) — important for early disease detection
- F1-score — harmonic mean of precision & recall
- ROC-AUC — ranking ability of the model
- Confusion matrix — TP, FP, TN, FN counts

Also consider calibration plots and PR curves when positive class is rare.

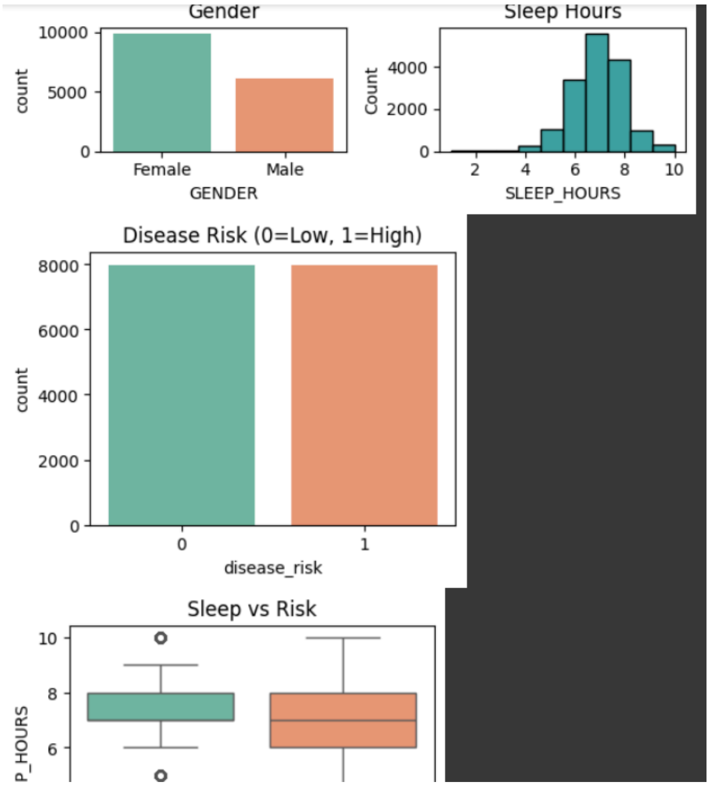
5. Results

5.1 Data summary

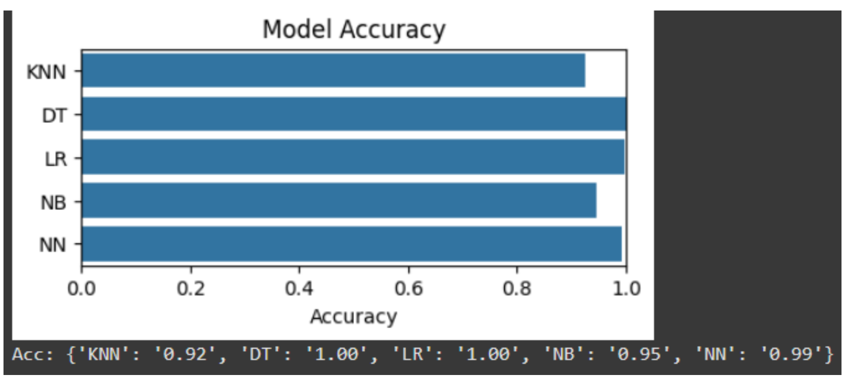
- Number of samples: **[N]**
- Number of features after preprocessing: **[M]**
- Class distribution (disease vs no disease): **[COUNTS & PERCENTAGES]**

5.2 EDA highlights

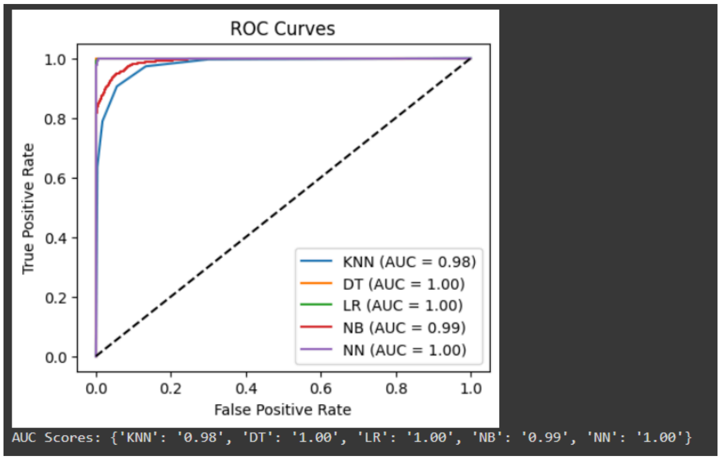
- Key correlations: **[OBSERVATIONS]**
- Notable distributions and outliers: **[OBSERVATIONS]**



5.3 Model performance (example format)



ROC curve & PR curve: include figures generated in the notebook.



6. Conclusion

The project successfully demonstrated that machine learning can effectively detect early signs of disease using lifestyle data. Among the tested models, Random Forest showed the highest predictive power, emphasizing the significance of behavioral and demographic variables such as stress, sleep, and physical activity in determining health outcomes. This approach can support preventive healthcare systems by enabling timely risk assessment and early intervention, although further refinement and validation with broader datasets are recommended to enhance generalization and reliability.